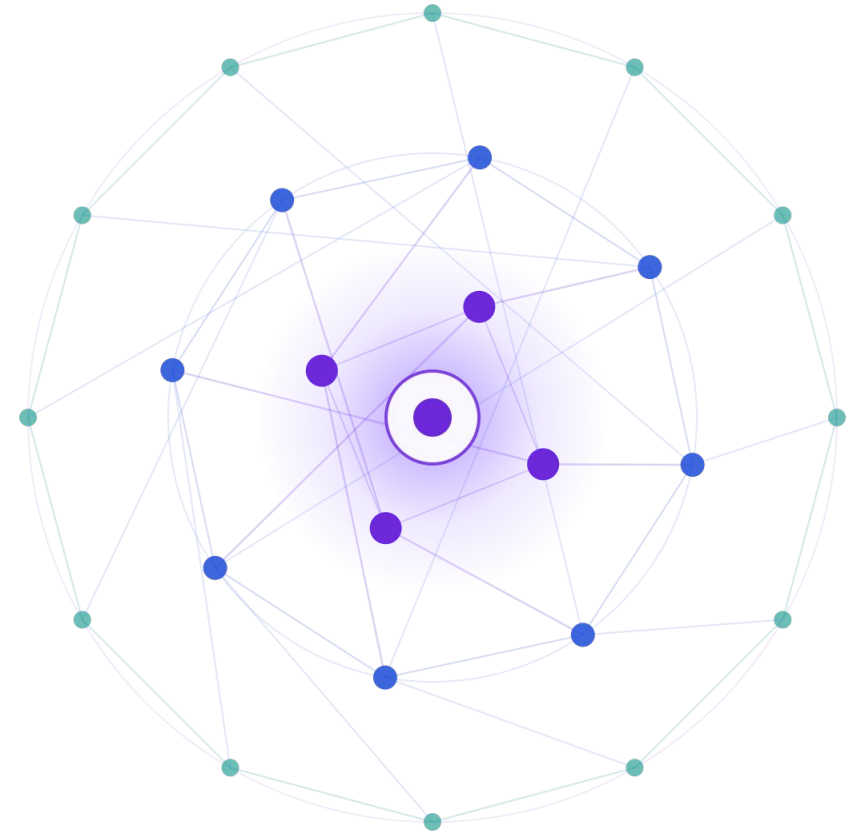


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - **Theme 3 — Teachable Voice Automation**
- Team Name - **Legacy**
- College Name - **Thapar Institute of Engineering & Technology**
- Member Name & Email 1- **Gurnoor Pannu — gpannu_be23@thapar.edu**
- Member Name & Email 2- **Durvish Khurana - dkhurana_be23@thapar.edu**
- Submission Github link - **github.com/gurnoorpannu/MYNA**



MYNA · Mimic Your iNteractions & Automate — show once, then just say it.

Theme 3 — Teachable Voice Automation

The problem: our phones make us repeat the same 8–15 taps every day, and no assistant can learn them.

1

Repetitive by design

Re-ordering the usual pizza, restocking groceries, re-buying a phone case: the same search, the same item, the same options — tapped by hand every single time.

2

Assistants can't learn

Bixby and Google Assistant only do what their makers built in. "My usual Domino's order on Zomato" is impossible unless the app ships an integration.

3

Automation is for experts

Tasker / MacroDroid / auto-clickers need rules or screen coordinates, and silently break the moment the app's layout changes or a pop-up appears.

What the theme asks for

Teach once

by voice + one demonstration

Replay by voice

with new values (item, qty, address)

Survive change

pop-ups, moved buttons, new screens

Ask when unsure

never a silent wrong guess

Stop at the boundary

zero taps on payment / OTP / login

Rules: AccessibilityService only · no app SDKs · no deep links · no hard-coded flows · flows must take parameters

Existing Solutions & Gaps

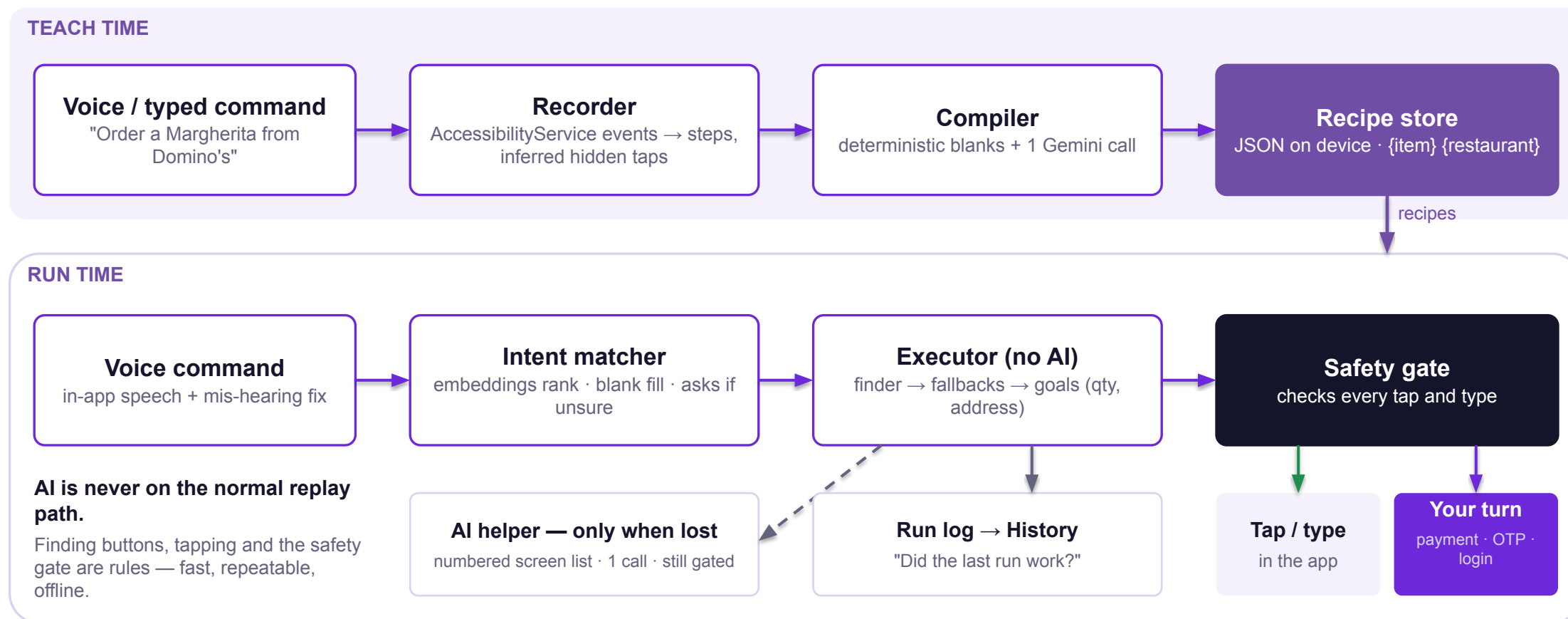
Approach	Learns a new app task from one demo	Voice with changed values	Survives UI changes	Hard stop at payment / OTP	AI cost per run
Bixby / Google Assistant	✗ built-in tasks only	✓ built-in intents	✓	✓	cloud NLU
Tasker · MacroDroid · auto-clickers	~ manual rules / coordinates	✗	✗ fixed positions	✗	none
Research record-replay (SUGILITE, VASTA)	✓	✓	~ text + OCR match	✗ no hand-off	none
LLM phone agents (AutoDroid, MobileGPT, AppAgent)	✗ re-plans every run	✓	✓	~ prompt-based	1 LLM call per step
MYNA (ours)	✓ one voice + tap demo	✓ blanks, qty, address	✓ 4-level finder + recovery	✓ rule gate, 0 AI	≈1 call per command, 0 on replay

The gap. Nobody combines one-shot teaching, parameterised voice replay and a hard safety boundary at near-zero AI cost.

Assistants can't learn new apps; macro tools break on the next UI change; LLM agents are slow, costly and non-deterministic — they re-think every tap, every time. Record-replay research proved the idea but never stopped itself at a payment screen or handled today's apps (web views, Compose lists, buttons drawn as images).

Our Solution & Architecture

Show once, then just say it. One demo becomes a recipe with blanks; a sentence replays it with new values.



Our Solution — learning a task from one demo

1 Capture every step

Merges all of the app's windows into one screen tree (UiNode).

Each tap gets an identity: label, id, sub-texts, neighbours, card anchor → a unique key.

Taps apps hide are inferred from what changed:

typed, then landed → search goal

sheet closed → confirm options (+ your choices)

keyboard gone → Enter pressed

web page changed / cart count up → "Add to cart"

Call screens, double taps and tap-then-Back are dropped as noise.

2 Generalise into a recipe

Blanks found without AI: spoken words that match what you typed, searched or tapped → {item}, {restaurant}.

Half-typed or typo'd searches are completed ("margerit" → Margherita).

1 Gemini call names blanks, writes a summary, paraphrases, a "why" per step and flags mistake steps.

Options you picked become habit defaults.

No key / no quota → rule-based fallback, still works.

THE LEARNED RECIPE

Order a {item} pizza from {restaurant} on Zomato

- 1 Open Zomato (clean start)
- 2 Search {restaurant} → open it
- 3 Search the menu for {item}
- 4 Tap ADD next to {item}
- 5 Confirm options — New Hand Tossed · Regular
- 6 Continue → checkout
- 7 Hand off at payment

Say "a farmhouse" → same recipe, new {item}

Our Solution — replay that recovers, safety that holds

Finding the right button on a changed screen

L1

Exact key the demo's id / label / child text

L2

Same element matched by neighbours and card anchor

L3

Fuzzy text Levenshtein ≥ 0.8 , singular/plural

L4

On-device OCR ML Kit reads text drawn as images



Bounded fallbacks close pop-up (X only) · open search · Enter · scroll
· best result

AI

Last resort Gemini picks from a numbered list — still gated



Stop with a reason closed restaurant · Hindi screen · no progress
in 30 s

Safety gate — rules only, zero AI

Payment screens UPI / card / wallet / EMI choices, pay buttons

Credentials password, OTP and secure fields

Final commit Place order, Pay, Send order to business

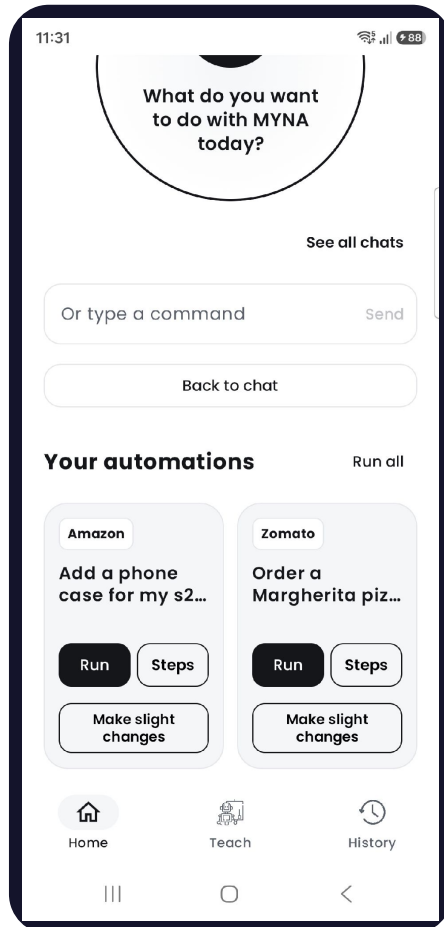
Login screens sign-in and account walls

GatedActor is the only code path that can tap or type — every action is checked first.

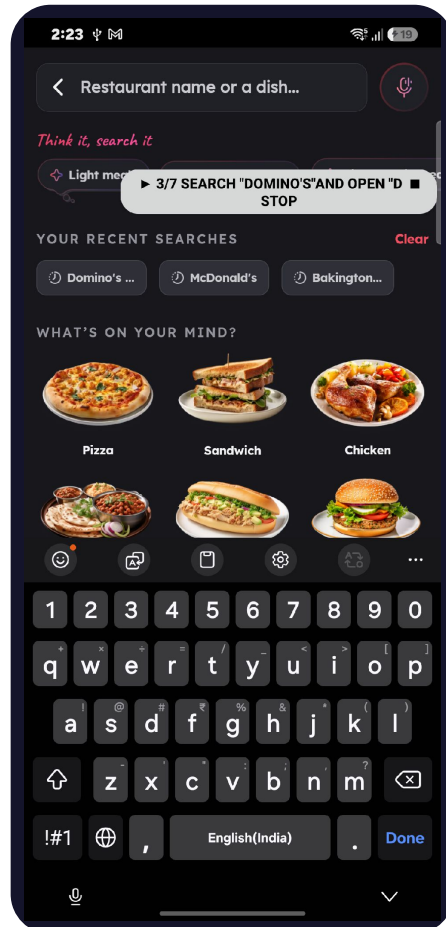
One exception: choosing a saved address on the checkout screen ("deliver to work").

Tuned on real screens: cart offers, a Wallet tab and colour radios are not payment.

Demo & Product Walkthrough



Home: mic + saved automations



Replay: live progress, ■ Stop

- 1 Teach**
Say "Order a Margherita pizza from Domino's on Zomato", tap Show me once and just do it. The 🛑 stops the demo at payment.
- 2 Learned**
MYNA says "Learned: ..." and shows every step with its why. Mistake taps are left out.
- 3 Just say it**
Tap the mic: "get me 2 farmhouses from dominos, deliver to work". A chat confirms what it heard.
- 4 Replay**
MYNA opens Zomato, searches, adds, sets quantity and address — asking only if unsure.
- 5 Your turn**
Zero taps on payment. "Did the last run work?" → one-sentence report; History shows each step.

► **Demo video (≤ 5 min):**
drive.google.com/file/d/1DHrpGlnpbme7hSHadp0A3DoQJjBCN2yX

Tools and Tech Stack



App

Kotlin 2.2 · Jetpack Compose (Material 3) · minSdk 30 (Android 11+) · Poppins UI



Vision

ML Kit on-device text recognition — reads buttons drawn as images



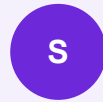
Voice

Android SpeechRecognizer in-app, partial results, biased to recipe words · TextToSpeech replies



Quality

172 JVM tests (full replays on fake Zomato & Amazon built from real screen dumps) + 8 on-device tests



Screen control

AccessibilityService: window trees, events, gestures, screenshots, key events



Language AI

Gemini Flash (JSON schema, temp 0, 1 retry) · gemini-embedding-001 for intent ranking



Data

kotlinx.serialization JSON recipes & run logs on device · private apps never sent to AI



Build & tools

Android Studio · Gradle AGP 9 · adb debug hook for dumps/replays · GitHub

Impact & Use Cases

8–15 → 1

taps become one sentence

0

taps on payment, OTP or login

≈1

AI call per voice command — 0 on the replay path

Any app

from the launcher — no SDK, no partner deal

Everyday re-orders

"My usual pizza", weekly groceries, medicines, a phone case — the same flow with a new item or quantity.

Accessibility

Older users and people with low vision or motor impairments say one sentence instead of hitting many small targets.

Hands busy

Cooking, carrying, multitasking: speak the task, check the result, press pay yourself.

Samsung ecosystem

A teachable layer for Bixby: users add their own skills on any app, with no integration from the app maker.

Privacy-first

Recipes and logs stay on the phone. Messaging apps (WhatsApp, SMS) are never sent to AI or saved as screens.

Innovation Highlights

1

Learns what apps hide

Web views, Compose lists and buttons drawn as images send no events — MYNA infers those taps from what changed on screen.

2

Blanks without guessing

{item}, {restaurant} come from matching your words to your taps — deterministic; AI only names and explains them.

3

Never-taught parameters

"2 margheritas, deliver to work" → taps + on the stepper and picks the Work address: steps never shown in the demo.

4

Asks instead of guessing

"Did you mean ... like last time?", "Which restaurant?", confirms 2+ changes, asks for a finger tap when an app refuses.

5

AI off the hot path

Replay, finding and safety are rules: repeatable and offline. Gemini is used 3 ways, each with a fallback.

6

Built for real phones

Speech mis-hearing fixes, slow-network waits, pop-ups closed by ✕ only, closed-restaurant and Hindi-screen stops.

Results — all 14 official test cases

ID	Test	Result
T1	Teach — food (Zomato)	✓ Pass
T2	Exact replay to payment	✓ Pass
T3	Paraphrase (2 wordings)	✓ Pass
T4	Slot: item (Farmhouse)	✓ Pass
T5	Slot: quantity (two)	✓ Pass
T6	Slot: address (work)	✓ Pass
T7	Screen change / pop-up	✓ Pass
T8	Teach — e-commerce (Amazon)	✓ Pass
T9	Cross-app slot + replay	✓ Pass
T10	Genuinely stuck → reports	✓ Pass
T11	Credential boundary	✓ Pass
T12	Unknown intent	✓ Pass
T13	Ambiguity → asks	✓ Pass
T14	Reporting the last run	✓ Pass

Every test has a screen recording linked in the README.

172 + 8

JVM tests (full fake-app replays) +
on-device tests

19.6 s

to teach the Zomato order (by hand: 34 s)

18.6–34.6 s

Amazon replay, hands-free, to the
payment step

1 + ≈1

AI calls: one per teach, about one per
command

Tested on Galaxy S25 Ultra and Nothing Phone (3a), Android 16.

Limitations — and how MYNA handles them

Limitation	What MYNA does about it
Zomato's "Add item" sometimes ignores accessibility taps	Reselects your options, tries twice, then asks for one finger tap and waits 30 s
Some taps send no event while teaching	Inferred from screen changes; replay also re-finds the result or ADD by name
On-device OCR reads Latin script only	Stops on Hindi screens with a clear reason instead of tapping blindly
Gemini free tier $\approx$ 20 calls/day, sometimes 429/503	Rule-based fallbacks keep teaching and replay working offline
Replay can be slower than by hand (41 s vs 34 s)	The win is hands-free and error-free; waits only while pages load
No phone-call pause, no re-teach from one step	Run ends with a reason in History; steps can be deleted or re-taught
Run button replays the taught values	Say or type quantity / address in the chat for those runs

By design: replay always stops at payment — MYNA never places an order or pays.

What's Next

Next

- Cross-app replay: teach on Amazon, run on Myntra / Flipkart (universal search, pick-result and cart goals)
- Re-teach from any single step
- Pause and resume around phone calls
- Ask for missing values even when you name only the app ("order a pizza on Zomato")

Soon

- On-device model (Gemini Nano) for blank filling and recovery — fully offline
- Hindi and Indic OCR + speech
- Share and import recipes between users
- Highlight each element just before it's tapped

Vision

- "Hi Bixby, my usual order": MYNA as the teachable layer of Bixby
- Triggers by time or place ("every Sunday 8 pm")
- Multi-app chains: order food, then WhatsApp the ETA
- One recipe, every similar app — a shared library of learned tasks

The goal: every Galaxy user teaches their phone once — and never taps the same flow again.

Brownie Points — what sets MYNA apart

Bonus criteria



Discards unnecessary touches

Call screens and the dialer are never recorded; double taps and tap-then-Back are dropped; AI flags leftover mistakes → "I left out 2 steps that looked like mistakes."



Mid-flow parameter clarification

A missing blank is recognised and asked: "Which restaurant? Last time it was Domino's" — then the flow continues with your answer.



Generalisation across similar apps

Search, pick-the-best-result and checkout are app-agnostic goals, proven on Zomato and Amazon; running an Amazon recipe on Myntra is next.

3 bonus criteria · 2 fully met, 1 foundation built

Built on research, tuned on real apps

Paper	What we took → what we added
SUGILITE (CHI '17)	demo + voice teaching → payment hand-off, hidden taps
VASTA (Samsung, IUI '20)	3-level finder + OCR → thresholds re-measured on Gemini
AutoDroid (MobiCom '24)	numbered screen list → used once, as last resort
MobileGPT (MobiCom '24)	sub-task recipes → universal search / pick goals
OS-Kairos (ACL '25)	act vs ask by confidence → asks for missing blanks
Pop-up attack studies	never tap a pop-up's action → close by X only

Extras: speech mis-hearing fixes · chat history like a messenger · volume key silences MYNA · WhatsApp catalog orders stop before Send.

Checklist — updated on public GitHub

Y	Working prototype code — public GitHub repo	github.com/gurnoorpannu/MYNA
Y	README with reproducible setup instructions	build, run, AI key, tests — README.md
Y	Demo video, max 5 minutes	drive.google.com/file/d/1DHrpgInpbme7hSHadp0A3DoQJjBCN2yX
Y	Presentation file (PPT or PDF)	this deck
Y	One video per official test case (T1–T14)	linked in the README test table
Y	Architecture and known limitations docs	docs/architecture.md · docs/limitations.md

Thank you

MYNA · Team Legacy · github.com/gurnoorpannu/MYNA

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India